

$$ds^2 = -\left(1 - \frac{R_s}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{R_s}{r}} + r^2 d\Omega^2$$

$\underbrace{d\Omega^2}_{d\theta^2 + \sin^2\theta d\varphi^2}$

$$\left(\frac{dr}{d\lambda}\right)^2 + \left(\frac{L^2}{r^2} + \epsilon\right) \left(1 - \frac{R_s}{r}\right) = E^2 \quad (1)$$

$$L = r^2 \frac{d\varphi}{d\tau}$$

$$L=0 \Rightarrow \left(\frac{dr}{d\tau}\right)^2 + \left(1 - \frac{R_s}{r}\right) = E^2$$

↳ Massive object: $\epsilon = 1$.

$$\frac{dr}{d\tau} = \sqrt{E^2 - 1 + \frac{R_s}{r}}$$

$$\tau = \int d\tau = \int_{r_0}^{R_s} \frac{dr}{\sqrt{E^2 - 1 + \frac{R_s}{r}}}$$

At $r = r_0$, start at rest:

$$\frac{dr}{d\tau} = 0, \text{ Hence: } 1 - \frac{R_s}{r_0} = E^2$$

$$E^2 - 1 = -\frac{R_s}{r_0}$$

$$\tau = \int d\tau = \int_{r_0}^{R_s} \frac{dr}{\underbrace{\sqrt{\frac{R_s}{r} - \frac{R_s}{r_0}}}_{\text{this is velocity}}}$$

→ finite amount of time to reach the singularity

$$\frac{dr}{d\tau} = \sqrt{\frac{R_s}{r} - \frac{R_s}{r_0}}$$

$$\frac{dr}{d\tau} = \frac{dr}{dt} \frac{dt}{d\tau} = \frac{dr}{dt} \frac{E}{1 - \frac{R_s}{r}} = \sqrt{\frac{R_s}{r} - \frac{R_s}{r_0}}$$

$$\Rightarrow dt = \frac{E dr}{\left(1 - \frac{R_s}{r}\right) \sqrt{\frac{R_s}{r} - \frac{R_s}{r_0}}}$$

$$t = \int_{R_s}^{r_0} \frac{E dr}{\left(1 - \frac{R_s}{r}\right) \sqrt{\frac{R_s}{r} - \frac{R_s}{r_0}}}$$

$\uparrow$
 $r = R_s + \delta$

$$\rightarrow = \int_0 \frac{E d\delta}{\left(1 - \frac{R_s}{R_s + \delta}\right) \sqrt{\dots}}$$

$$= \int \frac{E d\delta}{\left(1 - \frac{1}{1 + \frac{\delta}{R_s}}\right) \sqrt{\dots}}$$

$$= \int \frac{E d\delta}{\delta \sqrt{\dots}}$$

diverges

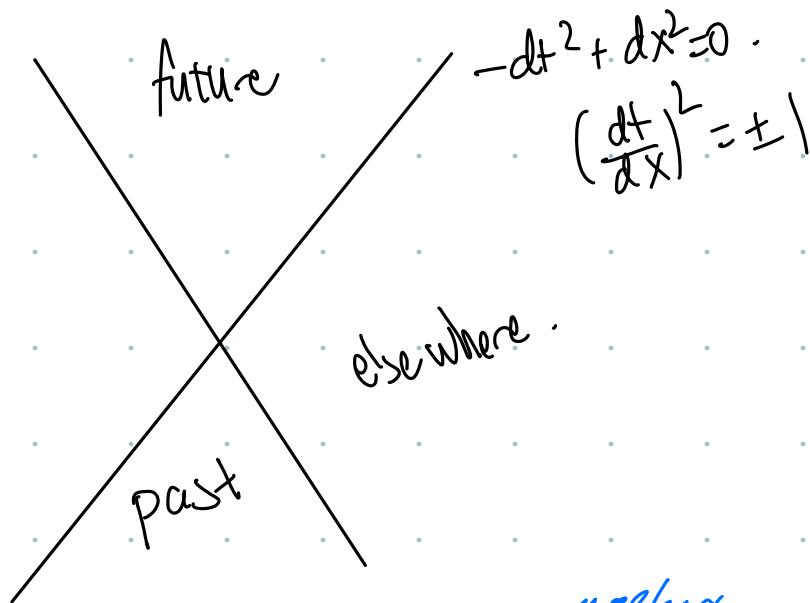
$\hookrightarrow \frac{1}{x}$ diverges?

$\rightarrow \infty$?

$\nearrow_{\infty}$ coordinate time -

Proper time maximized along
geodesic.

$$\int dr = \int \sqrt{g_{\mu\nu} \frac{dx^\mu}{dx} \frac{dx^\nu}{dx}} dx$$



$$ds^2 = -\left(1 - \frac{R_s}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{R_s}{r}} + r^2 d\Omega^2$$

→ neglect

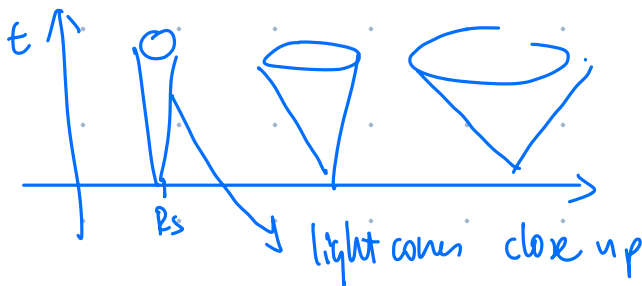
$d\Omega^2 = d\theta^2 + \sin^2\theta d\varphi^2$

$$r = R_s$$

$$0 = -\left(1 - \frac{R_s}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{R_s}{r}}$$

$$\left(\frac{dt}{dr}\right)^2 = \frac{1}{\left(1 - \frac{R_s}{r}\right)^2}$$

$$\frac{dt}{dr} = \pm \frac{1}{1 - \frac{R_s}{r}} \quad r \uparrow \quad \frac{t}{r} \downarrow$$



due to the fact that
 we are using coordinates
 that are not appropriate

consider
 the
 twin
 paradox

$$r^* = r + R_s \ln \left| \frac{r}{R_s} - 1 \right|$$

$$dr^* = \frac{dr}{1 - \frac{R_s}{r}}$$

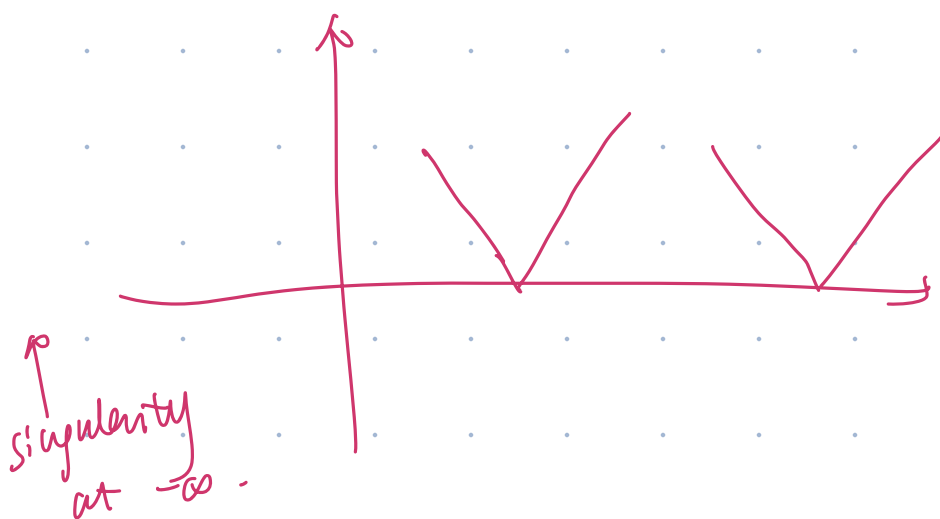
$$ds^2 = -\left(1 - \frac{R_s}{r}\right) dt^2 + (dr^*)^2 \left(1 - \frac{R_s}{r}\right)$$

$$= \left(1 - \frac{R_s}{r}\right) (-dt^2 + (dr^*)^2)$$

Not quite satisfied:

when $r = R_s \rightarrow r^* = -\infty$.

but you get nice light cones.



kind of get
the silly coordinates
iden.

$$dr^* = \frac{dr}{1 - \frac{R_s}{r}}$$

$$V = t + r^*$$

$$dv = dt + dr^*$$

$$\begin{aligned} ds^2 &= \left(1 - \frac{R_s}{r}\right) \left(-(-dv - dr^*)^2 + dr^{*2}\right) \\ &= \left(1 - \frac{R_s}{r}\right) (-dv^2 - 2dvdr^*) \\ &= -\left(1 - \frac{R_s}{r}\right) dv^2 + 2dv \frac{dr}{1 - \frac{R_s}{r}} \left(1 - \frac{R_s}{r}\right) \\ &= -\left(1 - \frac{R_s}{r}\right) dv^2 + 2dvdr. \end{aligned}$$

$$g_{\mu\nu} = \begin{pmatrix} -\left(1 - \frac{R_s}{r}\right) & 1 \\ 1 & 0 \end{pmatrix}$$

What does the light cone look like?

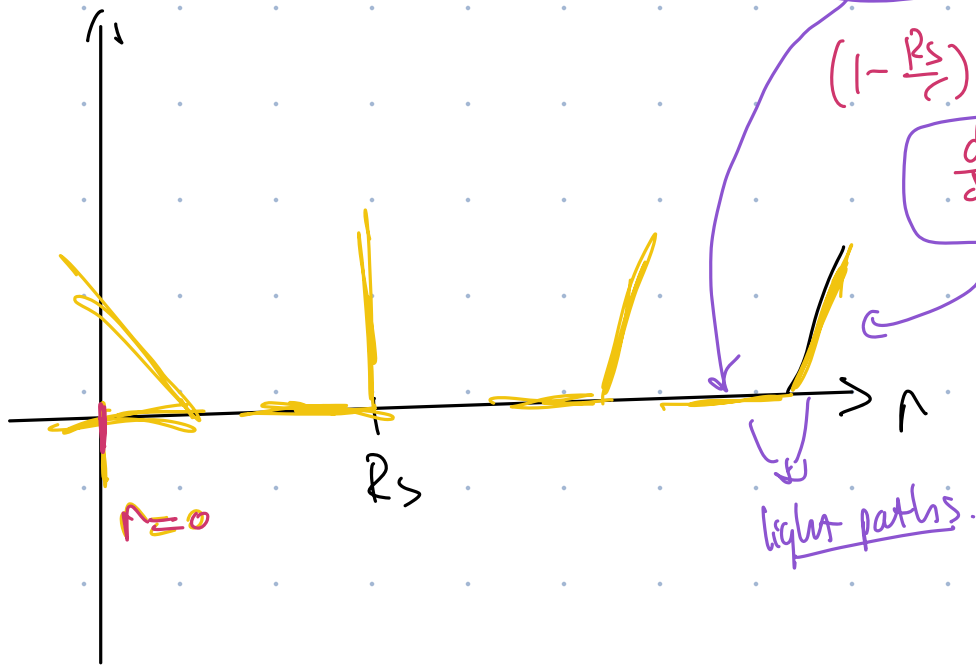
$$\left(1 - \frac{R_s}{r}\right) dv^2 = 2dvdr \rightarrow dv \left[\left(1 - \frac{R_s}{r}\right) dv - 2dr \right] = 0$$

$$dv = 0 \quad \text{or} \quad \frac{dv}{dr} = \frac{2}{1 - \frac{R_s}{r}}$$

either $dv = 0$ or

$$\left(1 - \frac{R_s}{r}\right) dv - 2dr = 0$$

$$\frac{dv}{dr} = \frac{2}{1 - \frac{R_s}{r}}$$



$$\left. \begin{aligned} u &= \sqrt{\frac{r}{R_S} - 1} e^{\frac{r}{2R_S}} \cosh \frac{t}{2R_S} \\ V &= \sqrt{\frac{r}{R_S} - 1} e^{\frac{r}{2R_S}} \sinh \frac{t}{2R_S} \end{aligned} \right\} \begin{aligned} u &= \sqrt{1 - \frac{r}{R_S}} e^{\frac{r}{2R_S}} \cosh \frac{t}{2R_S} \\ V &= \sqrt{1 - \frac{r}{R_S}} e^{\frac{r}{2R_S}} \sinh \frac{t}{2R_S} \end{aligned} \quad \left. \begin{aligned} r &= 0 \Leftrightarrow \\ v^2 - u^2 &= 1 \end{aligned} \right\}$$

$R_S = 2G_N M$

$ds^2 = -(1 - \frac{R_S}{r}) dt^2 + \frac{dr^2}{1 - \frac{R_S}{r}} + r^2 d\Omega^2$ (Schwarzschild coordinates)

Kruskal - Szekeres coordinates

$$\left. \begin{aligned} u &= \sqrt{\frac{r}{R_S} - 1} e^{\frac{r}{2R_S}} \cosh \frac{t}{2R_S} \\ v &= \sqrt{\frac{r}{R_S} - 1} e^{\frac{r}{2R_S}} \sinh \frac{t}{2R_S} \end{aligned} \right\} r > R_S$$

$$\left. \begin{aligned} u &= \sqrt{1 - \frac{r}{R_S}} e^{\frac{r}{2R_S}} \sinh \frac{t}{2R_S} \\ v &= \sqrt{1 - \frac{r}{R_S}} e^{\frac{r}{2R_S}} \cosh \frac{t}{2R_S} \end{aligned} \right\} r < R_S$$

$ds^2 = \frac{4R_S^3}{r} e^{-r/R_S} (-dv^2 + du^2) + r^2 d\Omega^2$

$u = \frac{1}{\sqrt{1 - \frac{r}{R_S}}} e^{\frac{r}{2R_S}} \sinh \frac{t}{2R_S}$
 $v = \frac{1}{\sqrt{1 - \frac{r}{R_S}}} e^{\frac{r}{2R_S}} \cosh \frac{t}{2R_S}$
 $\frac{v}{u} = \coth \frac{t}{2R_S}$

